JASPER BUNTINX

MECHANICAL ENGINEERING AT TEXAS A&M UNIVERSITY

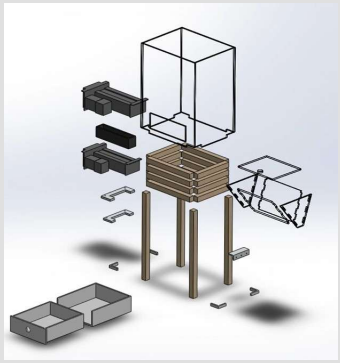
CONTACT

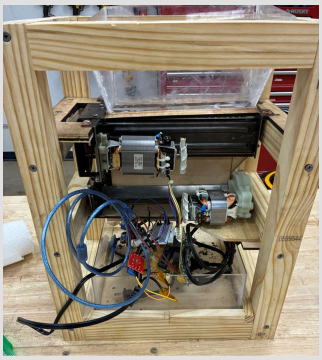
jasper_buntinx@tamu.edu
(737)-247-8938
Austin, TX, USA

ECOFIL-FILAMENT RECYCLER (ONGOING STARTUP)

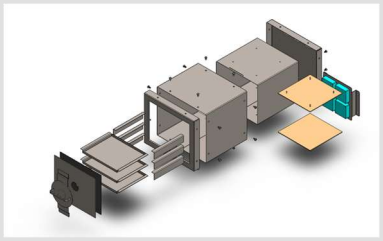
Description- Designed a multi-stage filament recycling system (shredder, dehydrator, and extruder) in SolidWorks, ensuring precise component mating and functionality while applying GD&T principles.

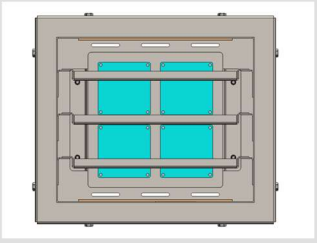
PHASE 1 - PLASTIC SHREDDER



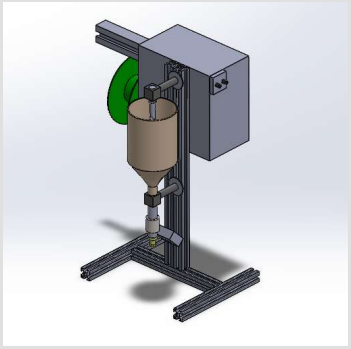
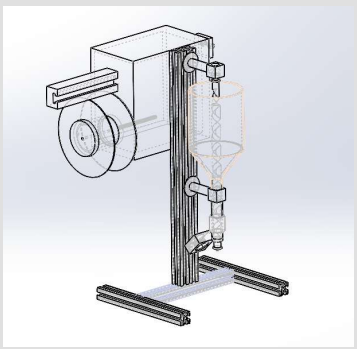
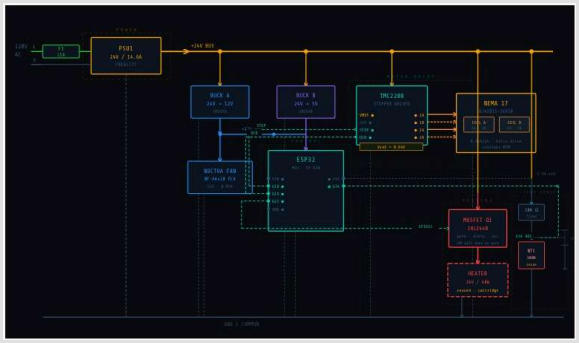



PHASE 2 – FILAMENT DEHYDRATOR





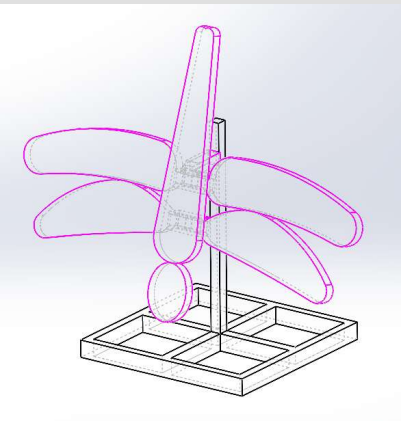

PHASE 3 – FILAMENT EXTRUDER

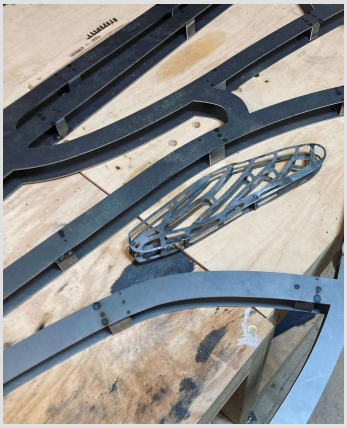




AUSTIN CREEK SHOW – LARGE-SCALE METAL SCULPTURES

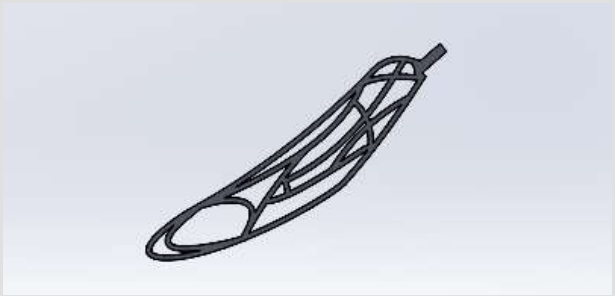
Description- Created initial mockup designs, sample prototypes, and final sculptures utilizing expertise in SolidWorks, fiber lasering, welding, metalworking, and circuitry.

MODELING, PROTOTYPING, MANUFACTURING

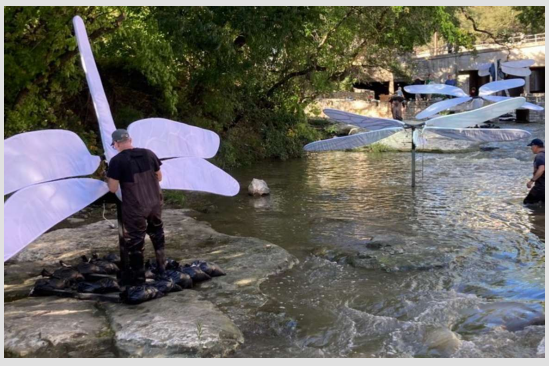






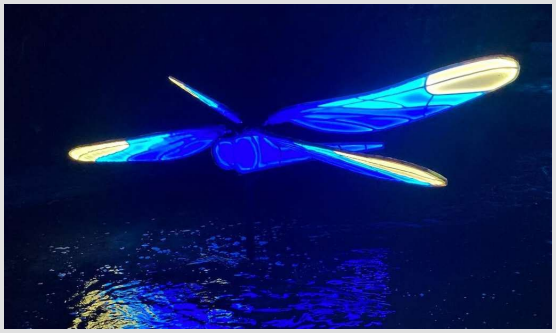
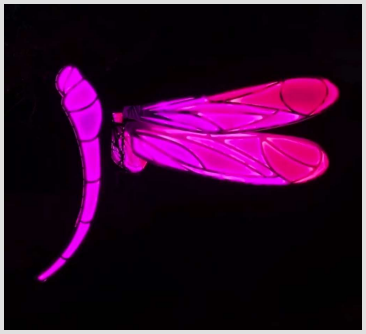


ONSITE + ASSEMBLY

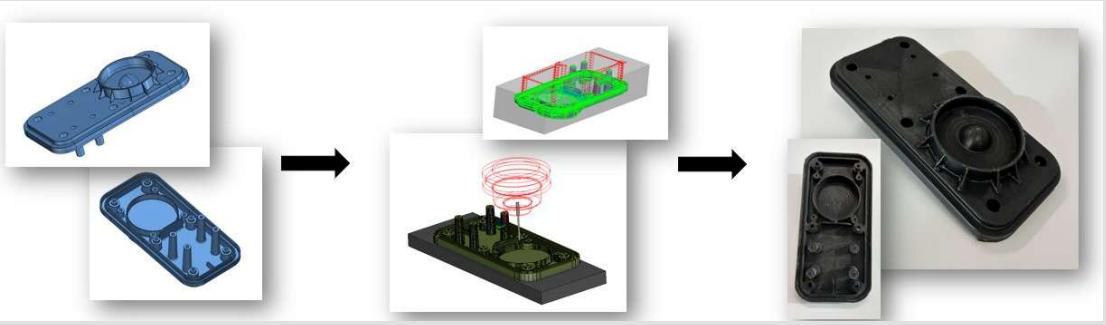


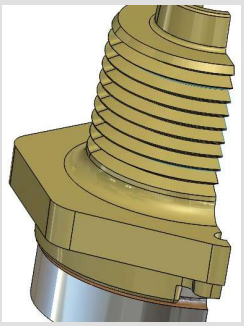
FINAL PRODUCT

ROCHESTER SENSORS–CNC MANUFACTURING

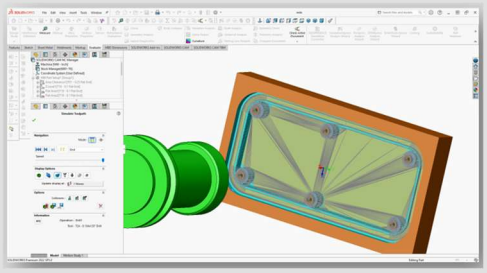
Description-Cut CNC cycle times 70-80% across 15+ production parts and qualified sensor assemblies past a 50-year equivalent field life. Also automated quality testing with URE5 and Keyence sensor.



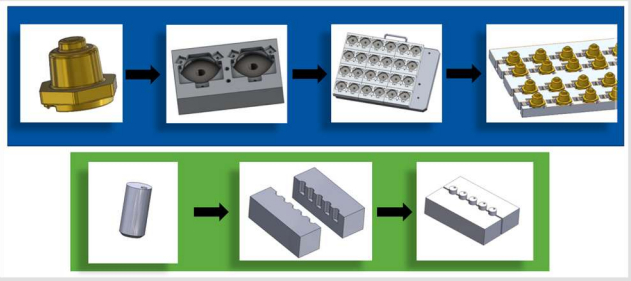


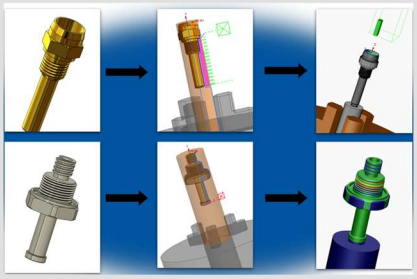




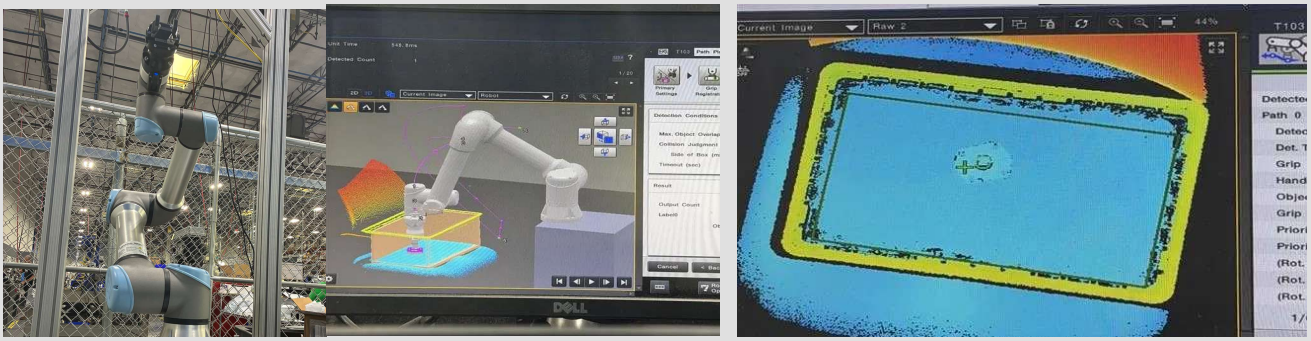








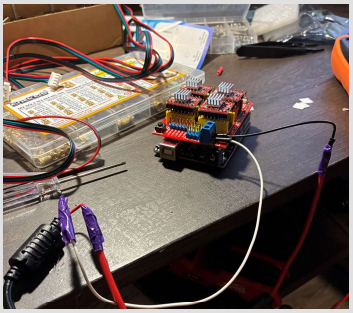
ROCHESTER SENSORS–QUALITY TEST AUTOMATION

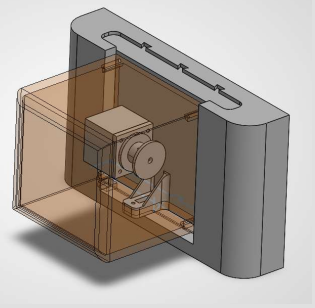


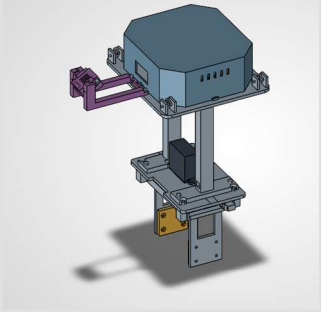
ROOM CLEANER–AUTONOMOUS LAUNDRY CABLE ROBOT

Description- Four ceiling winches fly a tendon-driven gripper on cables, guided by an overhead camera, to clear laundry off the floor.



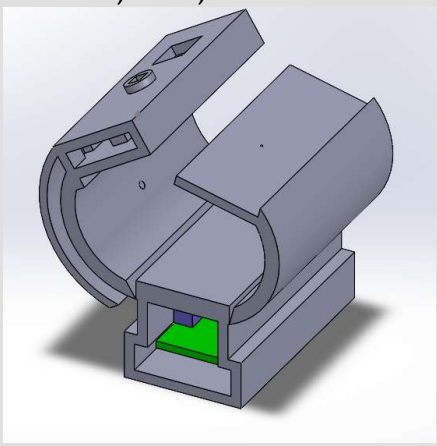
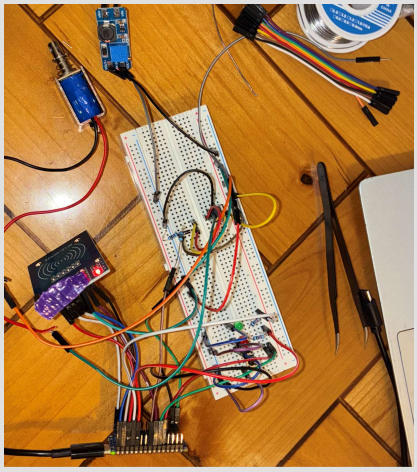
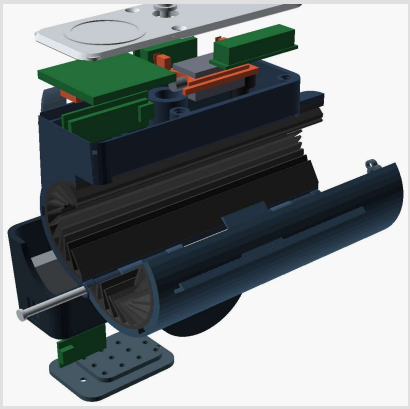


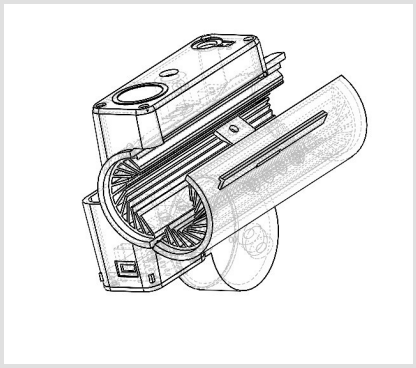




AUTOMATIC BIKE LOCK PROTOTYPE

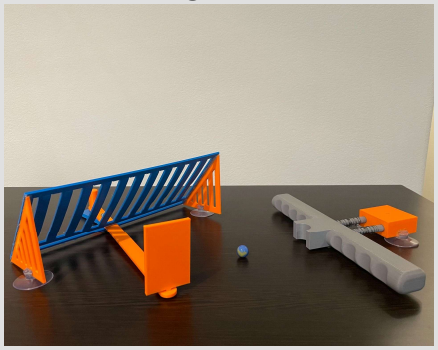
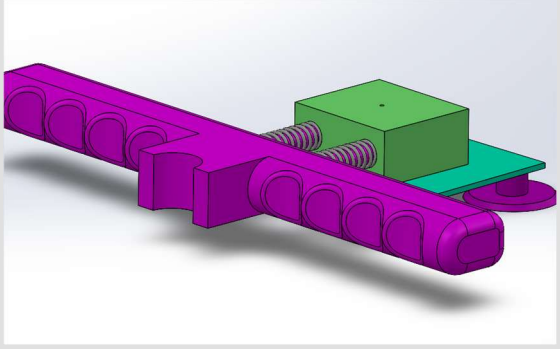
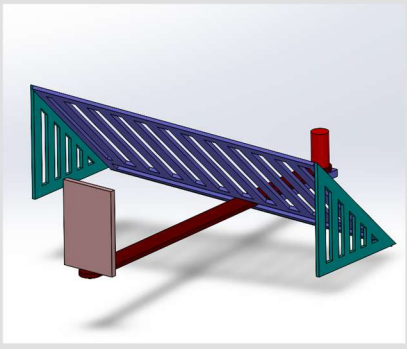
Description- Developing an automated locking mechanism integrating Arduino Nano, RFID scanner, solenoids, LEDs, and audio feedback.




SOCCER SHOTS–GAME PROTOTYPE

Description- Created CAD models (SolidWorks) for a multiplayer toy involving a spring-powered ball launcher and goalie mechanisms.

GAS TO ELECTRIC TRACTOR CONVERSION PROJECT

Description- Disassembled old gas-powered tractor, welding in new electric transaxle, taking out the engine, and replacing it with a battery system to make the tractor fully electric.

